

BlueKoi: Agility through unpowered turns in a fish-inspired robot

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Abstract—Bio-inspired, fish-like robots are promising platforms for collecting data within sensitive environments, due to the relatively less disruptive nature of oscillatory flapping propulsion. However, current fish-inspired robots often cannot match the maneuverability of their biological counterparts. This study aims to close the gap by expanding on our previous work BlueKoi, a koi- and tuna- inspired robot that captures the benefits of body flexibility with only two actuators. We introduce a rapid maneuver for angular displacements, named the unpowered turn, which utilizes a change of the robot’s morphology to turn without active tail flapping. We perform a parametric analysis investigating the impact of other control variables on the turning performance, including the time spent coasting in the unpowered turn, the speed during the initial forward motion phase, and the angular speed in which the head turns in the turn. We explore how combinations of maneuvering strategies can achieve more complex trajectories that would be relevant for obstacle avoidance within a prescribed larger area in a real-world environment. Furthermore, we extend our simplified, flat-plate model to the unpowered turn, demonstrating that the maneuver can be described through hydrodynamic equations.

Index Terms—bio-inspired, robotics, ocean vehicles

I. INTRODUCTION

Coral reefs, seagrass beds, and kelp forests are critical marine environments that demand frequent and widespread field observations, particularly data that requires close-range monitoring. Propeller-based craft have greatly expanded data collection capabilities, yet disruptive propeller wash limits how closely they can approach animals and delicate natural structures [1]. Bio-inspired, fish-like robots offer a promising alternative, as their oscillatory flapping motion has a lower risk of being destructive, posing less danger to surrounding organisms and habitats. Despite recent advancements, however, robotic fish remain considerably less maneuverable than their biological counterparts. This gap stems in part from morphological constraints: robots must house their power sources, electronics, and actuators within a rigid body, reducing the overall flexibility that is central to fish-like agility [2].

Recent fish-inspired robots utilize different mechanical systems to move through water, making strides to close the gap between real fish and robotic ones. Linking actuators such as servos in series [3], [4] allows for control of both

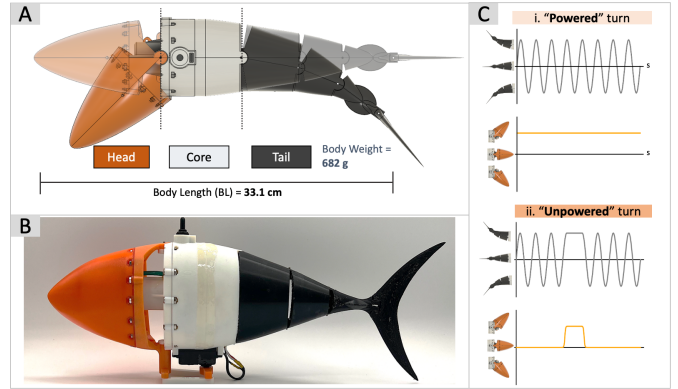


Fig. 1. (A) The design of the robot, along with the subsections (head, core, tail), and demonstrating maximum positions for head and tail. (B) Photo of the robot. (C) Head and tail positions for a **powered** turn (i) and **unpowered** turn (ii).

the amplitude and frequency of the tail, enabling dynamic maneuvers. However, this introduces mechanical complexity and may reduce maximum operating frequency. Hydraulically actuated and other soft propulsors have shown promising results, even seeing deployment in real environments [5], [6], but their manufacturing process is difficult to scale over a low-cost constraint. Other robotic fish [7], [8] utilize a switching electromagnet to move the caudal fin. While compact and capable of maneuvers without utilizing body bending, these actuators are limited in the maximum thrust that can be produced.

Our previous work introduced BlueKoi [9] (Fig 1A, B), a koi- and tuna-inspired robot designed for exploring how body morphology can enhance maneuverability, and eventually, for field deployment. Despite containing only two actuators, BlueKoi executes maneuvers that exploit full-body flexibility while remaining mechanically simple, in contrast to previous works.

This study expands upon our previous work on BlueKoi by presenting a novel 2D maneuver, inspired by fish “c-start” [10], that fully utilizes BlueKoi’s unique morphology to execute highly tunable turning. We experimentally quantify the turning behavior and the driving variables, and we show that these results can be correlated to a modified version of our

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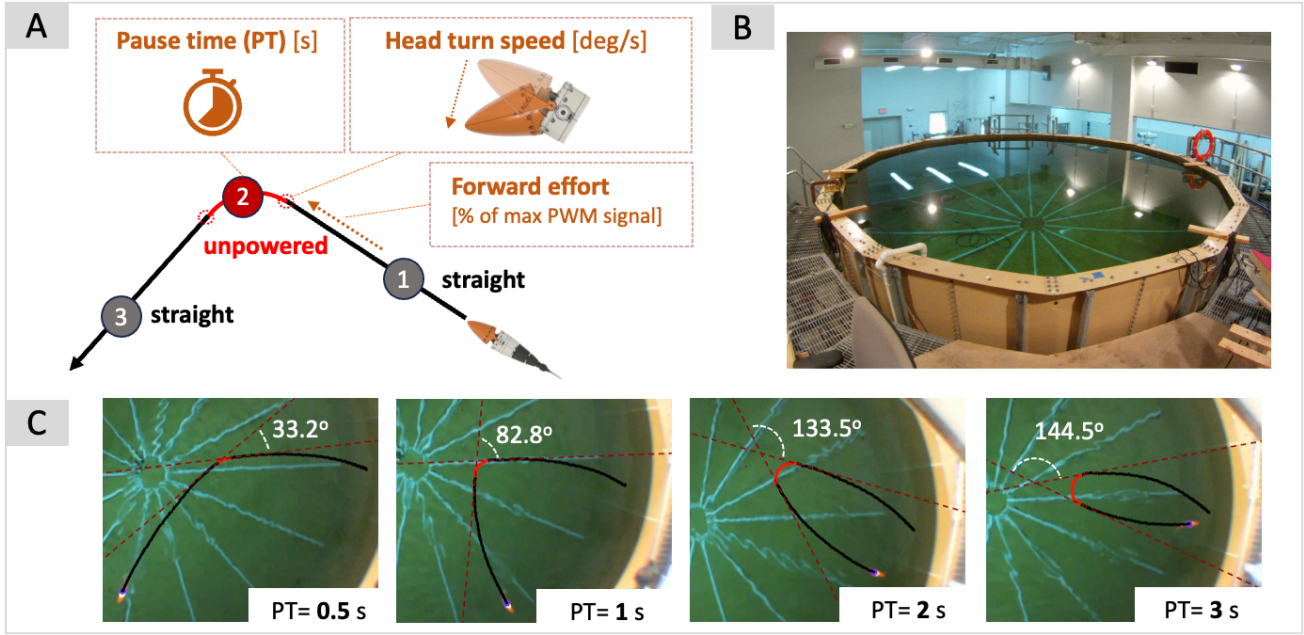


Fig. 2. (A) The variables in the unpowered turn experiments, which all consist of swimming in a straight line, turning, then swimming in a straight line again. (B) The pool used in the experiment. (C) Example trajectories of the unpowered turn, from a preliminary experiment. The black segments represent straight line swimming, while the red portion represents the unpowered turning. The dashed red lines represent the average heading two seconds prior and after the turn.

previous data-driven model, demonstrating BlueKoi’s potential as a low-cost and scalable platform for ocean exploration.

II. BACKGROUND

A. Turning in 2D

Fish-inspired robots that rely solely on the caudal fin for maneuvering are non-holonomic in their motion. As a result, turning in 2D can be non-trivial. One approach is to employ an asymmetric tail beat, where moving the tail from one direction differs in speed than the other [2]; however, this can be challenging to implement mechanically, requires precise sensing and control of the tail’s position, and may limit the total generated thrust from the tail. Instead, BlueKoi performs turns by leveraging its unique ability to rotate both its head and tail to break the body’s spatial symmetry. One maneuver the BlueKoi is capable of is a **“powered”** turn, wherein the head is actuated to one side to steer the robot while the tail continues to flap symmetrically (Fig. 1C(i)). This creates unequal drag across the left and right sides of the robot, resulting in a wide circular turn [9]. However, the powered turn results in a large minimum turn radius and is unsuitable for rapid maneuvers such as last-second collision avoidance.

A new and more agile maneuver we demonstrate in this study is the **“unpowered”** turn, during which the head and tail both are rapidly actuated to one side and held in place (Fig. 1C(ii)). The sudden, rapid change to an asymmetric “c” shape coupled with an abrupt and loss of thrust results in a momentum-driven tight angular displacement, or partial turn.

B. Robot Design

BlueKoi’s primary design consists of three subsections: the head, the core, and the tail, as shown in Fig 1A. The head serves as a battery hold, the core contains the primary electronics and actuators, and the tail is a flapping propulsor, based off of the Tunabot Flex [11]. The tail flapping motion is driven by a DC motor. The head pivots on an axis, controlled by a servo. A potentiometer mounted at the motor shaft provides angular information about the tail’s position. More on the robot design can be found in the previous work [9].

III. EXPERIMENTS

A. Methodology

In order to better understand the characteristics and limits of the unpowered turn, we quantify the maneuver through trajectory tracking experiments in a large pool (Fig. 2B). The trajectories are executed in three phases: 1) straight forward swimming, 2) unpowered turning, 3) straight forward swimming in a new direction (Fig. 2A). All phases utilize open-loop control; a program executes each phase based on a set amount of time that each phase should last. The initial forward swimming generates momentum for the unpowered turn. During this first segment, the head is held centered while the tail provides thrust; we set the **forward effort** [% of maximum PWM signal], which effectively controls the forward speed. The maximum possible PWM signal provided to the motor is 100%, equivalent to using 100% of the available battery voltage to power the brushed motor; for these experiments, we used 75% as the baseline to conserve energy. The corresponding frequencies and speeds can be found in

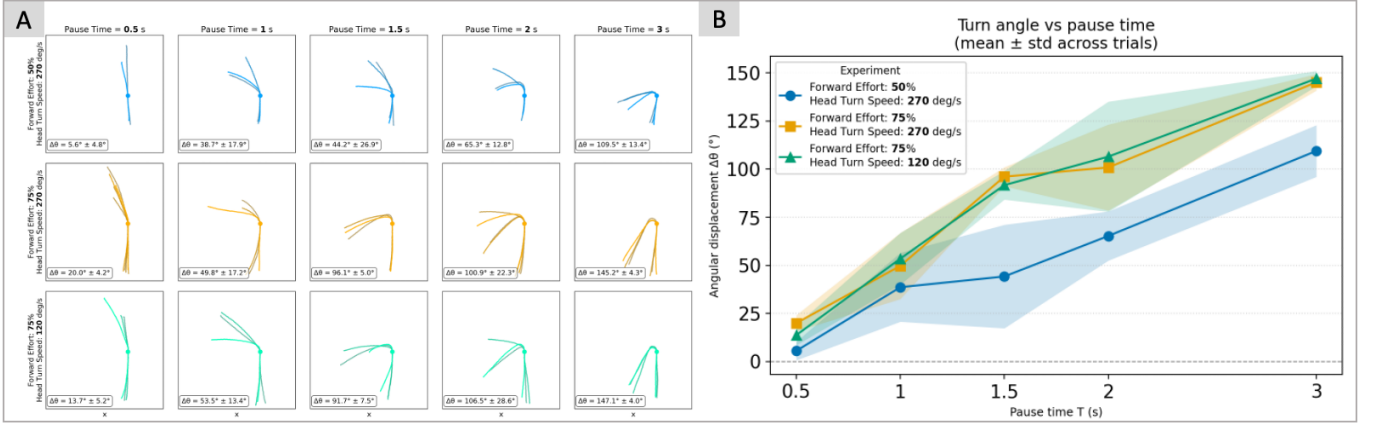


Fig. 3. (A) Each trial’s trajectory given a set of parameters. The trajectories are rotated such that each trial’s pre-turn average heading is pointing north. (B) The average angular displacement given each set of parameters. The colored bands represent the standard deviation for the corresponding experiment.

our previous work [9]. During the turn phase, both the head and tail are held at the leftmost angle from center. The angular speed which the head turns from 0 degrees center to 45 degrees left during the beginning and end of the unpowered turn is determined by the **head turn speed** [deg/s]. The time spent passively turning in the unpowered turn phase, is known as the **pause time (PT)** [s], after which the robot resumes forward swimming.

For the experiments, we vary the following parameters: pause time (0, 0.5, 1, 1.5, 2, 3 seconds), forward effort (50%, 75% of maximum effort), and head turn speed (270 deg/s, 120 deg/s). A total of 45 trials are collected over 15 parameter sets, or experiments. BlueKoi’s motion is recorded through the overhead camera, and the trajectory is backed out through DeepLabCut [12].

B. Results

From these experiments, we observe how pause time, forward effort, and head turn speed impact the total angular displacement achieved during the turn. We can quantify the turning by determining an approximate change in heading between the two straight motion paths. By taking the average heading two seconds before the unpowered turn and two seconds after, we can use the difference to determine the total angular displacement of the turn.

The trajectories from each trial are shown in Fig 3A, with each trial trajectory rotated such that average heading of the two seconds prior to the unpowered turn are aligned with north on the page. The straight line swimming paths appear slightly curved due to a combination of camera distortion and small bias in the robot’s steering, due to the use of open loop control for straight line swimming. The average angular displacement is shown in Fig 3B, plotted as a function of the pause time, with the errors represented as the colored bands.

Across all tested configurations of forward effort and head turn speed, increasing pause time generally increases the angular displacement achieved. When entering the unpowered turn at a slower speed, such as going from 75% of the maximum

forward effort to 50%, the angular displacement is generally decreased given the same pause time. However, changing the head turn speed given the same forward effort does not appear to significantly affect the angular displacement.

C. Discussion

We theorize that the unpowered turn occurs through two physical mechanisms. First, as the head rapidly moves through the water at the beginning of the turn, the body and tail rotate due to recoil and in reaction to the head’s sudden angular displacement. Then, though this initial turning moment may start the turn, BlueKoi’s head and tail are now positioned toward the left; this configuration creates unequal drag on the sides of the robot, causing it to turn as linear momentum is converted into rotational momentum without active flapping.

Increasing pause time allows for more time for the body to rotate before returning to straight line swimming. However, past a certain point, increasing the pause time does not increase the overall angular displacement due to the body’s deceleration from moving through a fluid.

Decreasing the forward effort while entering the unpowered turns also decreases the forward speed, therefore the drag that acts upon the body to induce a turn is also lessened, as drag scales off of speed. Thus, moving slower before turning translates to less rotation achieved.

Reducing the head turn speed appears to minimally affect the average extent of the unpowered turn. This shows that the rotation is primarily driven by drag caused by the asymmetry of the body as the linear energy is converted into rotational energy and less so by the sudden rapid movement of the head through the water.

In general, variability between the trials is expected, due to the nature of BlueKoi’s underactuated tail. Though the drive mechanism for the tail stops the flapping shaft in the leftmost position, there may be slack between each joint along the tail. Thus, each trial may have a slightly different tail position, which affects the resulting turn. In addition, there may be underlying fluid movement from waves bouncing off

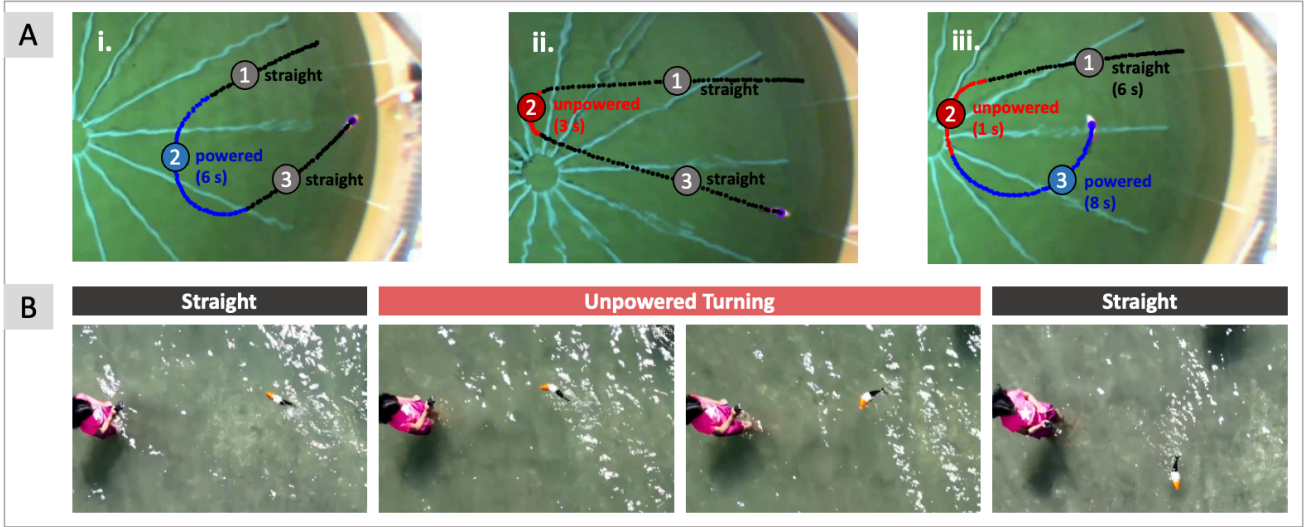


Fig. 4. (A) i. Powered turn, over 6 seconds, preceded by and followed by straight line swimming segments. ii. Unpowered turn, over 3 seconds. iii. A combination of both straight, unpowered, and powered turning. (B) The unpowered turn, with a pause time of 2 seconds, on a shallow beach in Fisherman's Cove, New Jersey.

the pool walls, which would have the greatest impact during the unpowered turn where the robot is not actively swimming.

D. Comparison to powered turn

Compared to the powered turn, the unpowered turn requires less time to achieve approximately equal angular displacement as an equivalent powered turn, as shown in Fig. 4A(i)(ii). The powered turn, over the course of 6 seconds, requires more time and space to turn around; on the other hand, the unpowered turn is quicker and tighter, requiring about only 3 seconds.

However, the unpowered turn requires forward momentum prior to the maneuver; furthermore, because the rotational momentum is damped over time, the robot may not be able to reach larger angular displacements, while the powered turn can. These maneuvers can be used in combination though, to overcome the disadvantages of each one, giving rise to more complex motions such as the one shown in Fig. 4A(iii).

E. Field experiments

Although the experiments were conducted primarily in a controlled lab environment, we show that the unpowered turn can still be used as a maneuver in more unpredictable environments, such as a nearshore environment (Fig. 4B). In our real-world trial, we allow the robot to swim forward for about 5 seconds before pausing and turning for 2 seconds, then returning to straight line swimming. From the snapshots of one example experiment, we can observe that the robot is able to achieve an approximate 90 degree left turn, while battling waves. However, compared to the lab experiments, the overall achieved angular displacement is less. This may be due to the fact that the movement of the waves dampen the rotational momentum faster than in a still pool. Nevertheless, we demonstrate the unpowered turn's potential in real-world and turbulent environments.

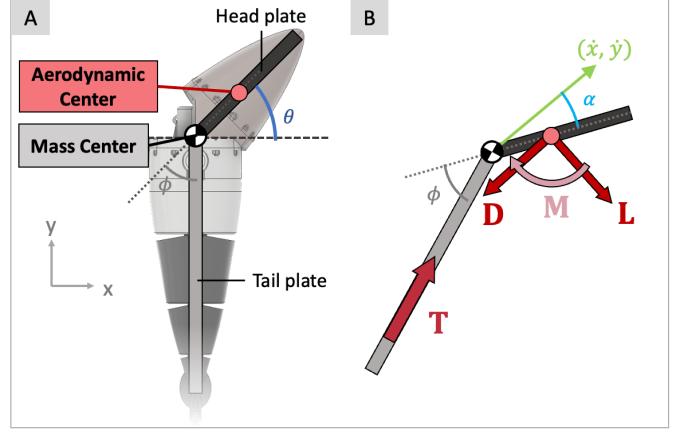


Fig. 5. Adapted from [9]. (A) The model of the robot as two flat plates, with the center of body coordinates inline with the head plate. (B) The forces and moments acting on the body, along with the angle of attack (α) and the body velocity $(\dot{x}, \dot{y})$, the latter also representing the direction of flow that determines the directions of lift and drag.

IV. MODEL

A hydrodynamic model allows us to predict the trajectory and to build a foundation for controls. The model described in the previous paper [9] is a simplified, flat-plate representation of the robot (Fig. 5), and utilizes the 2D hydrodynamic forces acting on the bodies to describe a trajectory. We extend the simplified flat plate model to the unpowered turn case in order to better understand the enabling physics.

With this existing hydrodynamic model, we show that it is possible to represent the unpowered turn by changing the head angle; furthermore, the extent of the angular displacement can be controlled by changing the pause time in the simulation. This also demonstrates that the overall physics of the simpli-

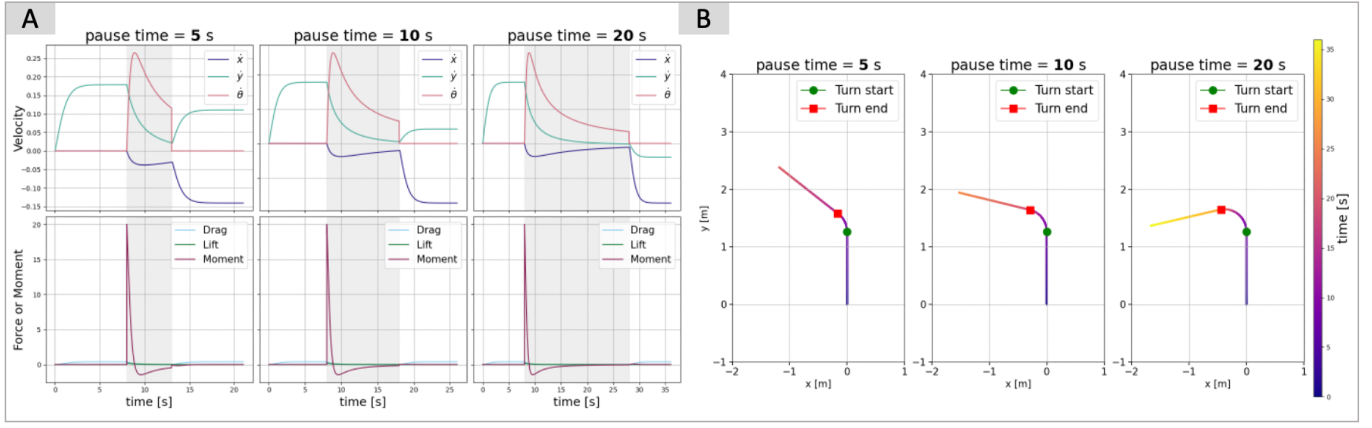


Fig. 6. (A) Top row: the kinematics over time of each pause time's simulation, with columns representing varying pause time. Bottom row: hydrodynamic forces and moments. (B) Trajectories from the simulation, with color corresponding to time.

fied model is able to capture maneuvers beyond the powered turn.

A. Equations of motion

The following subsection is adapted from [9]. The equations that describe the forces and their coefficients are:

$$\mathbf{T} = C_T \left(\frac{1}{2} \rho A f^2 \right) (\cos(\phi) e^{i\theta} + \sin(\phi) e^{i(\theta + \frac{\pi}{2})}) \quad (1)$$

$$\mathbf{L} = C_L \left(\frac{1}{2} \rho A \|\dot{x}, \dot{y}\|^2 \right) \frac{\dot{x} + i\dot{y}}{\|\dot{x}, \dot{y}\|^2} (i) \quad (2)$$

$$\mathbf{D} = C_D \left(\frac{1}{2} \rho A \|\dot{x}, \dot{y}\|^2 \right) \frac{\dot{x} + i\dot{y}}{\|\dot{x}, \dot{y}\|^2} (-1) \quad (3)$$

$$\mathbf{M} = C_M \left(\frac{1}{2} \rho A \|\dot{x}, \dot{y}\|^2 \right) - C_{Md} (\dot{\theta} |\dot{\theta}|) \quad (4)$$

The coefficients are defined as:

$$C_L = C_{L\alpha} \sin(2\alpha) \quad (5)$$

$$C_D = C_{D0} + C_{D\alpha} (1 - \cos(2\alpha)) \quad (6)$$

$$C_M = C_{M\alpha} (\sin(2\alpha)) \quad (7)$$

There are six unknown sub-coefficients: $[C_{L\alpha}]$ Lift, as a function of angle of attack (AoA), $[C_{D\alpha}]$ induced drag as a function of AoA, $[C_{D0}]$ profile drag, $[C_{M\alpha}]$ moment as a function of AoA, $[C_{Md}]$ moment damping, and $[C_T]$ thrust. The estimated coefficients from experimental data can be found in the previous paper.

The state space representation of the equations of motion is defined as:

$$S = \frac{d}{dt} \begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ m^{-1} \text{Re}(\mathbf{L} + \mathbf{D} + \mathbf{T}) \\ m^{-1} \text{Im}(\mathbf{L} + \mathbf{D} + \mathbf{T}) \\ I^{-1}(\mathbf{M} + \mathbf{N}l) \end{bmatrix} \quad (8)$$

where N represents the force perpendicular to the direction of the head plate, and l represents the distance between the center of mass and the aerodynamic center along the head plate. By integrating the forces and moments over time steps, we recover the trajectory of the simulation.

B. Extension to unpowered turn

To capture the behavior of the unpowered turn, we break the trajectory into three phases, in the same way as the experiments: straight line swimming, the unpowered turn, and then followed by straight line swimming. We set thrust to be $\mathbf{T} = 0$ during the unpowered turn phase, the head angle to be 45 degrees to the left while turning, and 0 degrees in the straight line phases. In these simulations, we use a time of 5, 10, and 20 seconds for pause time to demonstrate an approximate relationship between pause time.

We theorize that the primary mechanism driving the turn in the simulation is the sudden change in angle of attack when the head rotates to the left. This causes the moment term to become non-zero, based off the equations, which then creates an angular velocity. As the thrust remains zero throughout the unpowered turn though, the robot decelerates in both linear and angular speed, as shown by the kinematic and force plots in Fig 6A. We observe $\dot{\theta}$ increases rapidly at the start of the unpowered turn and then decays as the robot decelerates through the water, which is consistent with our observation for the experimental case. A simulated trajectory of three different pause times is shown in Fig 6B. We observe the trajectories increase in angular displacement as the pause time is increased, in a manner consistent with the experiments.

One important caveat is that the values of both time and the achieved angular trajectories in these simulations are not meant to be directly compared to physical experiments but rather to demonstrate that the model can give rise to behavior similar to the physical unpowered turn. Additionally, during the final straight line portion, we clamp the angular velocity or $\dot{\theta}$ to 0, which is explained in more detail further down.

C. Limitations of the current model

Despite capturing the qualitative behavior of the unpowered turn, the current model has several limitations. First, the coefficient values are not fully consistent with those used previously, likely because the unpowered turn is a more dynamic maneuver whose complexity is not fully captured by the simplified formulation. In particular, $C_{M\alpha}$ and C_{Md} differ from the values obtained in the original powered-turn experiments: the original $C_{M\alpha}$ is too weak to capture the impact of the rapid angle-of-attack change induced by the head motion, while C_{Md} is comparatively larger to better dampen the unpowered turn. Second, we found it necessary to clamp $\dot{\theta}$ to zero during the final straight-line phase, since the model otherwise retains residual angular velocity that is not fully damped out by the end of the turn; this suggests that the damping term is insufficient to capture the physics governing how the robot has little to no rotation at the end of an unpowered turn (see the pause time = 3 s case in the real experiments). Finally, the simulated turn does not achieve as much angular displacement as observed experimentally when comparing values proportionally, suggesting that an additional impulsive moment may need to be introduced at the start and end of the turn to account for the rapid swinging motion of the head and body through the water.

Despite these limitations, the model demonstrates that simply removing thrust and moving the head to the side is sufficient to produce a tight turn qualitatively similar to the experimentally observed unpowered turn.

V. CONCLUSION

Overall, we demonstrate that BlueKoi's distinctive body-bending configuration improves maneuverability and gives rise to a novel rapid unpowered turns, expanding safe deployment capability in unstructured and complex environments. Experimentally, we demonstrate that the extent of the turn can be controlled by either pausing for longer in the unpowered turn, or entering the turn at a faster speed; at the same time, the angular turn speed of the head does not significantly impact the performance of the turn. We demonstrate that this maneuver can be used in conjunction with the powered turn, and we also demonstrate its ability to function in more turbulent environments such as on a beach. Furthermore, we show that the simplified flat plate model is sufficient at capturing the behavior of the unpowered turn.

Utilizing an unpowered turn opens the possibility of better last-second collision avoidance; however, this maneuver requires forward motion before a turn. The unpowered turn also depends on the initial speed of the turn, which may be a limitation in certain scenarios. This study primarily demonstrates that using the body's rapid change in morphology in a robot with rigid parts can still enable agility; the unpowered turn can be used in conjunction with more complex maneuvers to achieve better agility in robotic fish.

The experiments presented in this paper relied on open-loop control; a natural next step is closed-loop control of the turn, guided by an onboard sensor such as an IMU. Learning-based

methods could also build on the intuition developed here to produce more adaptable and agile maneuvers. Finally, extending this motion into 3D is a promising direction, particularly in combination with diving strategies that also require forward motion, such as pectoral fins that enable pitching.

VI. ACKNOWLEDGMENTS

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VII. DIVERSITY STATEMENT

Inspired by a recent initiative to increase awareness and mitigate citation bias [13], we include a gender citation diversity statement using manually compiled data. Our references contain 8% woman (first author)/woman (last author), 50% man/woman, 8% woman/man, and 33% man/man. The statistics we report fail to include intersex, non-binary, and transgender people. We look forward to future work that could help us support equitable practices in science.

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